Data dictionary of `results/`

All result files are JSON (one file per benchmark problem unless stated otherwise). The field names and a few categorical labels were kept exactly as written by the analysis programs, which use Spanish names; this table gives their meaning. Fractions are given as numbers between 0 and 1 (e.g. `eajuste_med = 0.094` means 9.4 %) unless the name ends in `%`.

Common labels: `sistema` = problem; `segundos` = running time of that problem in seconds (on the machine that produced the file); `admisible` = admissible; `Stage 1` / `Stage 2` = stage of the selection; `Técnico` = technical failure (the nominal simulation fails); `salidas` = measured outputs; `estados` = state trajectories.

`selection/` and `settings/<variant>/` (analysis/selection.py)

Field	Meaning
`p`	number of estimated parameters of the PEtab problem
`S1`	subset returned by stage 1 (greedy selection by sensitivity energy)
`Rvar_S1_%`	R_var of the stage-1 subset (%)
`S1_cumple`	true if stage 1 reached R_var >= tau_R with at least 2 parameters
`cos_S1`, `erel_S1`	median cos(Delta) and e_rel of the stage-1 subset
`motivo_stage2`	why stage 2 was used ("stage 1 reaches R_var but is not admissible")
`etapa`	stage that produced the subset (`Stage 1`, `Stage 2`)
`S_antes_poda`	subset before backward elimination
`podados`	parameters removed by backward elimination
`S`	**final selected biological parameters**
`calibracion`	calibration parameters (scalings, offsets), always re-estimated with S
`Rvar_%`	R_var(S) (%)
`kappa`, `VIF`	condition number and largest VIF of the standardised columns of J for S
`cos_med`, `erel_med`	median over the scenarios of cos(Delta) and e_rel (no re-estimation)
`eajuste_med`	**median e_fit** (re-estimation of S and the calibration parameters)
`eajuste_cota`	upper bound e_rel used when e_fit was not computed (not used in the paper)
`eajuste_metodo`	present for the 8 problems where e_fit is computed to first order
`n_escenarios`	number of valid scenarios (of 15)
`admisible`	**admissible**:
`motivo`	reason when the problem could not be analysed
`T`, `etapa_paper`, `S_paper`	legacy fields of an earlier version of the analysis (state trajectories), not used

`external_validation/` (analysis/external_validation.py)

Field	Meaning
`division`	split: `condiciones nuevas` (held-out experimental conditions) or `extrapolación temporal (t ≥ c)` (later time points)
`n_condiciones`, `n_train`, `n_test`	number of conditions; training and test measurements
`S`, `|S|`, `n_calibracion`	subset validated, its size, number of calibration parameters

Field	Meaning
p_y	parameters that affect the outputs
locales_prueba, S_locales_prueba	parameters that only act in test conditions (excluded from the fits)
no_evaluable	why no training/test split was possible
A_chi2_test_red, A_chi2_test_full, A_chi2_test_theta0	real data: test chi2/n of the reduced model, of the full model (both re-fitted to the training data) and of the published theta0
A_chi2_train_red, A_chi2_train_full	real data: training chi2/n
B_chi2_test_theta0, B_chi2_test_red, B_chi2_test_full	synthetic data: test chi2/n (theta0 = signal-to-noise indicator)
B_epred_red, B_epred_full, B_epred_theta0	synthetic data: prediction error e_pred with respect to the known truth (1 for theta0)
B_error	failure of the synthetic experiment (simulation of the truth)
etapas_reanudadas	fits that were resumed from a checkpoint

robustness/ (analysis/robustness.py)

Field	Meaning
S, n_calibracion	subset (fixed) and number of calibration parameters
semilla, u	seed and the 15 random directions u(k) (15 x p), the same at every size
niveles	results per perturbation size ("1", "5", ..., "50" = +-1 %, ..., +-50 %)
niveles.*.n	valid scenarios
niveles.*.escenarios	indices k of the valid scenarios
niveles.*.eaj, erel, cos	e_fit, e_rel and cos(Delta) of every valid scenario
niveles.*.eaj_med, eaj_q25, eaj_q75, erel_med, cos_med	median and quartiles
patron	robustness pattern A-D (Results)
eajuste_metodo	linealizado (first order) or no lineal

single_parameter/ (analysis/single_parameter.py)

Field	Meaning
candidatos	number of candidates tested (up to 10 by energy and 10 by finite effect)
un_parametro	for each candidate: parametro, e_rel, e_fit, admisible
algun_parametro_solo_basta	true if some single parameter is admissible
mejor	best admissible single parameter j* (smallest e_fit)
mejor_efit_10, S_efit_10	median e_fit at +-10 % of {j*} and of the selected subset S
mejor_validacion	external validation of {j*} (same fields as external_validation)
mejor_validacion_calibracion	number of calibration parameters re-estimated with j* in that validation

nested_validation/ (analysis/nested_validation.py)

Field	Meaning
S, |S|, division, n_train, n_test	selected subset, its size, kind of split, number of training and test measurements
chi2_test_red_original	test chi2/n of the reduced model with the non-selected parameters at theta0
chi2_test_full_train	test chi2/n of the full model re-estimated on the training data (theta_train)
chi2_test_red_nested	test chi2/n of the reduced model with the non-selected parameters at theta_train
chi2_test_theta0	test chi2/n of the published estimate theta0
ratio_nested_vs_original	chi2_test_red_nested / chi2_test_red_original (above 1: worse without the theta0 anchor)

synthetic_seeds/<problem>/s<seed>.json (analysis/synthetic_seeds.py)

One file per problem and seed (42-61). Fields as in external_validation/ (keys B_*) for that seed: semilla = seed; p_y = free parameters; B_chi2_test_theta0, B_chi2_test_red, B_chi2_test_full; B_epred_red, B_epred_full (relative prediction error with respect to the truth; 1 = theta0); nfev_red, nfev_full = function evaluations of the fits; seg_red = seconds of the reduced fit; B_error = the simulation of the truth failed. B_*_full is absent for the four problems with more than 100 free parameters. RESUMEN.json and RESUMEN.md summarise all seeds (por_semilla: counts per seed; por_sistema: per problem, con_senal = seeds with signal, red_ok_senal = reduced e_pred <= tau_e among them, red_le_full_senal = reduced e_pred <= full).

random_subsets/ and calibration_only/ (analysis/baselines.py)

Field	Meaning
k	number of biological parameters of our subset (also used for the random subsets)
chi2_ours	test chi2/n of our subset (theta0 anchor), from nested_validation/
chi2_rand_median, chi2_rand_q1, chi2_rand_q3	median and quartiles of the test chi2/n over the random draws
chi2_por_semilla, n_seeds, n_validas	test chi2/n of each draw (seeds 0-19), number of draws, valid draws
chi2_vacio	test chi2/n with only the calibration parameters re-estimated (n_cal of them)
ratio_rand_vs_ours, ratio_vacio_vs_ours	baseline over ours (below 1: the baseline predicts better)
nota	sin calibracion, th0 directo = no calibration parameters: theta0 is evaluated directly

qr_selection/ (analysis/qr_selection.py) and brun_morris/ (analysis/brun_morris.py)

Field	Meaning
S_ours, S_qr, S_brun, S_morris	biological parameters selected by each criterion (same k)
chi2_ours, chi2_qr, chi2_brun, chi2_morris, chi2_perfil	test chi2/n of each subset (with the calibration parameters, theta0 anchor)
overlap, overlap_brun_ours, overlap_morris_ours	parameters shared with our subset
ratio_qr_vs_ours	chi2_qr / chi2_ours
detalle_brun	gamma = collinearity index of the selected subset, cumple_gamma = gamma <= 10, dmsqr_top = importance of the 15 candidates

Field	Meaning
detalle_morris	mu_star_top = mu* of the 10 most influential parameters, fallos_simulacion = failed elementary effects
segundos_brun, segundos_morris, tipo	seconds of each criterion (selection and fit); kind of split

profile_likelihood/ (analysis/profile_likelihood.py)

<problem>.json:

Field	Meaning
p_libres, n_bio	free parameters of the training fit; biological parameters profiled
n_identificables, n_no_identificables_practicos, n_planos	parameters identifiable (profile above Delta chi2 = 3.84 on both sides), bounded on one side, flat
S_perfil	the k parameters with the narrowest profile confidence interval
chi2_perfil, chi2_ours, chi2_qr, ratio_perfil_vs_ours	test chi2/n of the profile subset, of ours and of pivoted QR
overlap_perfil_ours, overlap_perfil_qr	parameters shared with our subset and with the QR subset
coste_min_train	chi2 (sum of squares) of the full training fit used as the minimum
rondas_reajuste	times the full model was re-estimated because a profile point improved the minimum
dchi2_minimo_perfiles	smallest Delta chi2 among the profile points after the last re-estimation (below -1: residual improvement)
puntos_fallidos	profile points whose fit failed (Delta chi2 > 1e4), excluded
perfiles	per parameter: clase (identificable, no identificable practico, plano), ancho_ic_log (width of the 95 % interval in log units; infinite if open), dchi2_max_izq, dchi2_max_der (largest Delta chi2 on each side)
no_evaluable	not profiled: more than 60 free parameters (PL_PMAX), or no training/test split

<problem>/perfil_<parameter>_<side>_<i>.json: one profile point (lado = side, despl = displacement of log|theta_j|, dchi2, nfev, theta = re-estimated parameter vector); <problem>/full_train.npz: minimum (theta, coste, rondas). logs/: summary lines of the runs (terminado = problem finished with seg_total seconds; reajuste = re-estimation of the minimum).

criteria_tests/ and criteria_tests_20pc/ (analysis/criteria_tests.py)

File names <problem>__<criterion>__adm.json (admissibility) and <problem>__<criterion>__sint_<seed>.json (synthetic), with <criterion> = ours, qr, brun, morris, perfil. criteria_tests/ uses perturbations of +-5 % (the level of the method); criteria_tests_20pc/, +-20 % (admissibility only).

Field	Meaning
S	biological parameters of the subset tested
eajuste_med, erel_med	median e_fit and median e_rel over the scenarios
admisible	min(eajuste_med, erel_med) <= 0.436 and at least 2 parameters (rule of the method)

Field	Meaning
n_escenarios, lineal	valid scenarios; true if e_fit was computed to first order (problems of config.FIRST_ORDER)
B_chi2_test_theta0, B_chi2_test_red, B_epred_red, semilla	synthetic experiment for that seed (as in external_validation/)
no_evaluable	sin subconjunto = the criterion has no subset for the problem; falla la simulación de la verdad = the truth could not be simulated

identifiability/ (analysis/identifiability_*.py)

Profile likelihood of the selected parameters in the reduced problem (S and the calibration parameters free, the other parameters at theta0, all measurements).

profiles/<problem>/minimo.npz: optimum of the reduced problem (theta, coste = chi2, rondas = re-fits after a profile point improved the optimum). profiles/<problem>/p_<parameter>_<+|->_<i>.json: one profile point (despl = displacement in log|theta_j|, log_theta, dchi2, theta = all parameters after the re-fit). profiles/<problem>.json:

Field	Meaning
chi2_por_gl, umbral_robusto	chi2 per degree of freedom at the optimum; threshold of the robust variant, 3.84 max(1, chi2/d.f.)
perfiles[].clase, clase_robusto	identificable (profile above 3.84 on both sides within [theta0/10, 10 theta0]), no identificable practico (one side), plano (neither)
perfiles[].ic_izq_log, ic_der_log	distance (log) at which the profile crosses the threshold on each side (inf = no crossing)
perfiles[].minimo_en_limite, puntos_fallidos	sides without valid points (optimum at a bound); points whose simulation failed

local/<problem>.json: kappa_SC_n, VIF_SC_n_S (condition number and VIF of the unit-norm, uncentred columns of S and the calibration parameters at theta0), par_mas_parecido_con_S (most similar pair of columns and their cosine). certificado*.csv: one row per selected parameter with the quantities of the Lean theorem ProfileIdentifiability.practically_identifiable_grad: I_j (profiled information), gamma (norm of the projection of the residual onto the range of the sensitivity matrix), eps_j (largest nonlinear remainder at the profile points), tstar, tstar_lineal (Wald half-width), d_menos, d_mas (distances to the bounds), certificado_lineal, certificado_local (hypotheses satisfied with epsilon = 0, or with epsilon evaluated at the points up to the first crossing). summary.md: the counts quoted in the paper.

timing_complete/ (analysis/selection.py run from scratch)

parallel/, single_core/: the complete result of selection.py (fields as in selection/) run without cached results, with the sensitivity computations on 16-64 cores or on one core; the run time is segundos. parallel_corrected_code/: the eight problems of config.FIRST_ORDER with the corrected program (version 1.1); their subsets and eajuste_med are identical to selection/. The runs shared a 128-core server, so the times are upper bounds.

robustness_independent/ (analysis/robustness.py with ROBUST_INDEP=1)

Same fields as robustness/, plus direcciones ("independent at every size") and, in every size of niveles, semilla (seed of the directions of that size: 42 at +-5 %, 42 + 1000 i at the i-th other size). The top-level u are the directions of seed 42; the scenarios of the other sizes are not the rows of u.

`synthetic_signal_a<amplitude>/` and `synthetic_seeds_a<amplitude>/` (analysis/synthetic_seeds.py)

One file per problem and seed (`<problem>/s<seed>.json`), as in `synthetic_seeds/`, with the amplitude a of the truth theta* = theta0 (1 + a u) in the folder name (005 = 0.05, 020 = 0.20, 050 = 0.50) and in `amplitud`. New fields (also in `synthetic_seeds_a005/`, `_a050/`):

Field	Meaning
`senal_rms_test`, `senal_rms_train`	root mean square of y(theta*) - y(theta0) per test / training measurement, in units of sigma
`senal_norma_test`, `ruido_norma_test`	norm of y(theta*) - y(theta0) on the test measurements, and sqrt(n_test) (norm of unit noise)

The `synthetic_signal_*` folders contain only these fields (no fits). `synthetic_seeds_a*` contain the same fields as `synthetic_seeds/` (`B_chi2_test_*`, `B_epred_red`, `B_epred_full`, ...).

`absolute_error/` (analysis/absolute_error.py)

Field	Meaning
`n`, `n_escenarios`	number of measurements; valid scenarios at +-5 %
`e_fit_med`	median e_fit over the scenarios (from `robustness/`, +-5 %)
`cambio_rms_sigma_med`	median of `|Delta y_full| / sqrt(n)`, the change of the full model per measurement in units of sigma
`error_rms_sigma_med`, `error_rms_sigma_max`	median and maximum of `e_fit * |Delta y_full| / sqrt(n)`, the error that remains after re-estimation, in units of sigma
`escenarios`	the same quantities for every scenario k

`settings_extra/` (analysis/selection.py –variant)

Same fields as `selection/`, for Lang and Raimundez, in folders named as in `settings/`; `README.md` lists the variants that did not finish.

Other files

File	Content
`simulator_check/simulator_check.jsonl`	one line per problem: `mediciones`, `condiciones`, `p`, `p_y`, `err_rel_mediana` and `err_rel_max_vs_simulations` (relative difference with the reference simulations of the benchmark), `nota` (why no comparison was possible)
`cost/simulation_times.json`	`p`, `t_sim` (seconds per simulation), `modo` (`salidas` or `estados`); Smith: `t_sim_medido` and `nota` (scaled from a second machine)
`crauste_blowup.json`	`nominal` and `un_parametro_1%` (24 one-at-a-time changes): `existe_en_[0,28]` (the solution exists on the measured horizon), `t_final` (time reached), `efectores_max`, `t_max`; `tolerancias`: number of failures (`fallos_*`) for each integrator tolerance
`sbml_to_lean/<problem>.json`	report of the translation to Lean: sizes (`n_estados`, `n_theta`, `reacciones`), checks done in Python before writing the Lean file, `lean` (file written), `teorema_final`, `traducido`

File	Content
sbml_to_lean/translation_fidelity.json	for each model: err_rel_max (largest relative difference between the vector field of the Lean file and libRoadRunner), err_theta, err_x0, err_por_punto

Second review (version 1.5)

selection_fixedscen/, selection_train_fixedscen/ (analysis/selection.py)

Same fields as selection/, plus:

Field	Meaning
outputs	all (all measured outputs) or train (rows of the training set only)
fixed_scenarios	true: a scenario in which the simulation with the subset fails counts as e_rel = inf; first-order e_fit = min(e_rel, projection) per scenario
n_rows	number of outputs used by the selection
monotonia	monotonicity log: subsets_evaluated, nested_pairs (pairs S < T of evaluated subsets), violations (S admissible and T not), subsets (each evaluated subset with its admissibility)

qr_selection_gks/ (analysis/qr_selection.py with QR_VARIANT=gks)

Same fields as qr_selection/ (S_qr, chi2_qr, chi2_ours, overlap, ...), plus variant = gks.

efit_nonlinear/<problem>/s<k>.json and efit_nonlinear/<problem>.json (analysis/efit_nonlinear.py)

Field	Meaning
escenario	scenario index (0-14)
erel, efit	relative error without and with nonlinear re-estimation of S and the calibration parameters
arranques	each starting point (theta0, escenario): error e, function evaluations nfev, optimiser status
descartado	why a scenario was discarded (full-model simulation failed, or change below the noise floor)
eajuste_med_nolineal, eajuste_med_primer_orden	median nonlinear e_fit and the first-order value of the selection
tasa_paso_nolineal, admisible_nolineal, n_escenarios_validos, segundos_total	pass rate, admissibility, valid scenarios, total CPU time

criteria_tests_fixedscen/, criteria_tests_20pc_fixedscen/, criteria_tests_sint_<family>/ (analysis/criteria_tests.py)

One file per problem, criterion (metodo: ours, train, brun, morris, perfil, qr, qr_gks, empty = k = 0, rand<i> = random subset i) and test (adm, real, sint_<seed>):

Field	Meaning
eajuste_med, erel_med, admisible	admissibility test as in the selection (admisible_sin_kmin: without the rule |S| >= 2)

Field	Meaning
tasa_paso, n_escenarios, escenarios_fijos, k, n_calibracion	pass rate, valid scenarios, fixed scenario set, subset size, calibration parameters
chi2_test, chi2_ours, ratio_vs_ours	real data: test chi2/n of the subset, of our main subset, and their ratio
familia, semilla, B_epred_red, B_chi2_test_theta0, senal_rms_test	synthetic test: family of truths, seed, e_pred, chi2/n of theta0 on the synthetic test data (>= 2: signal), true signal in noise units

conditioning_check.json (analysis/conditioning_check.py)

kappa_centred, VIF_centred (as in the selection), kappa_unit, VIF_unit (unit-length columns without centring, as in the Lean least-squares statements), kappa_unit_with_calibration.

identifiability/profiles_dense/, identifiability/profiles_selection_train_fixedscen/

Same format as identifiability/profiles/ (20 points per side, or the subsets selected on the training outputs).

identifiability/audit2/certificado_box[_r<r>].csv and ..._parametros.csv (analysis/identifiability_box.py)

One row per selected parameter, side (lado) and condition (condicion: abs = absolute remainder on the tail, rel = relative remainder): I_j, gamma, norma_r0, d_lado (distance to the bound of the region), pasa_muestra (condition holds on the sample), t (tail start), valor_final (epsilon or eta after the adversarial search), pasa_final. _parametros.csv: pf = the hypotheses hold on both sides for parameter parametro in the region of radius r (10 if no suffix). certificado_train.csv: same columns as identifiability/certificado.csv for the training-output subsets.

Version 1.6

subsets_<criterion>/<problem>.json (analysis/prepare_criteria_subsets.py) and subsets_single/ (analysis/prepare_single_subsets.py)

S (subset of the criterion, or the best single parameter), calibracion (calibration parameters, those of our selection), admisible (marks the problem for the profiling programs; not the admissibility of the subset), origen (source file and key) or e_fit_5 and S_ours (single parameter).

identifiability/profiles_subsets_<criterion>/ and identifiability/profiles_subsets_single/

Same format as identifiability/profiles/. Only the problems whose subset differs from ours were profiled; the others use our profiles. identifiability/audit2/certificado_<criterion>.csv: same columns as identifiability/certificado.csv.

identifiability/criteria_identifiability.json (figures_tables/identif_criteria_tables.py)

common_problems (the 27 problems compared) and, per criterion, identifiable/selected parameters, all_identifiable, admissible_5, identifiable_and_admissible, identical_to_ours, cpu_h and the exact McNemar tests against our subsets (mcnemar_all_identifiable, mcnemar_both: problems in which only ours or only the other criterion has the property, p, p_holm).

robustness_subsets_single/, robustness_independent_subsets_single/

Same format as robustness/ for the best single parameter. partial/<problem>__L<size>.json: one perturbation size computed in its own process (ROBUST_LEVEL).

`kmin_comparison.json` (analysis/compare_single_vs_pair.py), `kmin_summary.json`, `kmin_times.json` (analysis/kmin_times.py)

Per problem: single parameter and pair; robustness (`pattern`, `admissible_up_to` = largest size with median e_fit <= tau_e, `pass_rate` = fraction of all scenarios with e_fit <= tau_e, `median_50`) with paired and independent directions; `adm20`; `synthetic` = (cases with e_pred <= tau_e, cases with signal); `identifiable` = (identifiable, selected). `summary`: sign tests over problems. `kmin_times.json`: elapsed time of the run and processor time per problem (seconds).